Automated Valuation Model for Vietnamese Landed Residential Houses

A Region-Segmented Reliability Report and Production Defense

AVM Project

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Abstract

This report documents an Automated Valuation Model (AVM) for Vietnamese landed residential houses, from raw-data preprocessing through the deployed model and its results. Its central argument is that a single dataset-wide accuracy number is the wrong lens for a real product: accuracy must be read *per region*, large errors must be *explained* rather than only minimised, and the system must *know when it cannot answer*. We show that the typical listing is valued to about 15% error, that the model is statistically unbiased, that its large errors concentrate in a thin population of mislabeled/anomalous listings (not ordinary homes), and that a confidence-tiered policy lets the product serve normal listings sharply while flagging anomalies. We close by diving into each region and, for the region that produces the most errors (Ho Chi Minh City), contrasting its worst error case against a normal case.

Contents

1 Data Preprocessing **2**

2 Data Used to Train the Model **2**

3 Results **3**

4 Interpretation for a Real Product **4**
4.1 The model knows when it is uncertain 4

5 Diving Into Each Region **4**

6 Error Cases per Region, and a Focus on the Worst **7**

7 Limitations and Next Steps **9**

Key Insights at a Glance

1. **Typical accuracy is strong.** The production model values the typical listing to $\sim 15\%$ error, with $\sim 59\%$ of estimates within 20%.

2. **The model is unbiased.** The alarming raw bias (+13.9%) is a mean-of-ratios artifact; the median bias is ≈ 0 overall *and in every region*.

3. **Big errors are bad labels, not bad predictions.** Large errors concentrate in listings with an implausibly low price/m^2 (mistyped / underpriced posts); the model actually predicts the sensible value and thus *catches* them.

4. **The model knows when it is unsure.** Its own interval width predicts its error (8% median APE on the most-confident fifth vs. 29% on the least), enabling an auto-serve / review / flag policy.

5. **Reliability is reported per region.** Da Nang, Khanh Hoa, Hai Phong are product-ready ($\sim$ 11–13%); **Ho Chi Minh** is the weak spot (23.7%) and the clearest next target.

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1 Data Preprocessing

Raw listings are scraped, heterogeneous, and noisy. The following steps turn them into a clean modeling table (implemented in `src/common.py` and the pipeline scripts):

1. **Parsing and typing.** Extract structured numeric fields (area in m^2, number of floors, frontage, road width, bedrooms/bathrooms, width, length, built year) and parse dates (post date, updated date) into a listing year.

2. **Address normalization.** Merge province/district spelling variants and abbreviations into a single canonical label — e.g. `tp.hcm`, `tphcm`, and `ho chi minh` collapse to one region; tone variants (`hoa`→`hoa`, etc.) are unified. This stops one real place from splitting into several location buckets.

3. **Property classification.** Assign each listing to the landed-residential family and one of seven subtypes (frontage, alley, normal, alley-frontage, townhouse-project, villa, shophouse) from text and structured signals.

4. **Text feature extraction.** Derive boolean flags from the free-text description (frontage, alley, business, car access, corner, villa, legal, project, urban area) plus text length and word count.

5. **Near-duplicate removal.** Collapse the same listing re-posted many times — matched on address, an area bucket (± 3 m^2), and a price bucket ($\pm$200M VND) — keeping only the most recent copy, so repeated posts do not dominate training.

6. **Suspicious-price auditing.** Flag and remove implausible prices (placeholder sequences, sequential/repeated digit runs, sub-500M values whose text says "ty"/billions), removing roughly 8,000 rows in the cleaned variant.

7. **Recency weighting.** Assign a training weight by listing year ($2025 \to 1.0$, older years progressively lower, undated $\to 0.5$) to down-weight stale prices.

8. **Leakage-safe outlier bands.** Compute local price-per-m^2 bands on the *training* data only, used to weight/robustify fitting without leaking test information.

2 Data Used to Train the Model

The cleaned data is split by a stratified random split on subtype:
Subtype mix (modeling rows): frontage 171k, normal 152k, alley 80k, alley-frontage 55k, townhouse-project 52k, villa 44k, shophouse 10k. The target is log(price); models are gradient-boosted trees (LightGBM, L1 objective) and predictions are back-transformed with $\exp(\cdot)$.

Stage	Rows
Cleaned listings	813,215
Modeling rows (landed residential, valid price/area)	704,370
Training set	563,496
Test set (held out)	140,874

Table 1: Data volumes through the pipeline.

3 Results

All models are evaluated on the same 140,874-row held-out test set.

Model	Median APE	Hit $\leq$20%	MAPE	Role
Global	21.96%	46.5%	44.0%	baseline
Subtype	20.45%	49.2%	42.6%	one model / subtype
Region	15.36%	59.0%	34.1%	one model / province
Hybrid (hard router)	15.39%	58.3%	36.1%	previous production
Shrinkage blend	**14.84%**	**59.5%**	**34.5%**	**current production**
Region×Subtype	14.01%	61.3%	32.1%	best raw (dense groups)

Table 2: Model comparison. *Median APE* is the typical listing's error and is the headline metric for a skewed target; *MAPE* (a mean) is inflated by a heavy right tail.

The production model. The **shrinkage blend** is the deployed model: it values *every* listing (unlike Region×Subtype, which only covers dense groups) and it improves on the previous hard router by -0.55 Median-APE points and $+1.14$ hit-rate points — a statistically solid gain (McNemar $p < 0.001$; net $+1{,}607$ listings moved inside the $\pm 20\%$ band).

The model is unbiased — read this before judging the numbers. The raw mean bias looks large ($+13.9\%$), but that is a *reporting artifact*. Because the model predicts the median, median(pred/true) $\approx$ 1.00; the robust bias views confirm calibration:

Bias measure (Region×Subtype)	Value
Raw mean bias	+13.9%
Median bias	−0.29%
10%-trimmed mean bias	+0.63%

Table 3: The model is centered; the raw mean is pulled up by a thin right tail (Section 6). Forcing the raw bias to zero would push Median APE from 12.8% to 16.8% — so we deliberately do not apply a naive bias correction.

Each valuation also carries a **P10–P90 prediction interval**; empirical coverage on the test set is 76.7% (target 80%). This interval is the foundation of the production defense.

4 Interpretation for a Real Product

Additional model tuning now moves the headline error by only fractions of a point. For a real product the useful question is not "what is the average error?" but:

> *For which listings, and in which markets, can we trust the valuation — and what does the system do when it cannot?*

This reframes the deliverable around three business-relevant claims, each defended with evidence:

1. **Reliability is segmented.** Accuracy is reported *per region*, because a national average hides that some markets are prediction-ready today and others are data-limited (Section 5).
2. **Errors are explained, not just minimised.** Large errors concentrate in a thin, *identifiable* population of anomalous / mislabeled listings, not ordinary homes (Section 6).
3. **Behaviour on new cases is defensible.** A normal incoming listing is valued sharply; the model *knows when it cannot* (via its own interval) and flags anomalies rather than emitting a confident wrong number (Section 4.1).

4.1 The model knows when it is uncertain

Grouping listings by the width of their own predicted interval — a signal available *before* the true price is known — accuracy falls monotonically:

Interval-width quintile	n	Median APE	Hit ≤20%
Q1 (narrowest)	28,175	8.05%	81.1%
Q2	28,175	11.48%	68.9%
Q3	28,174	15.47%	59.0%
Q4	28,175	20.24%	49.7%
Q5 (widest)	28,175	28.85%	38.7%

Table 4: The model's self-reported uncertainty is a strong, honest predictor of its real error.

This yields a three-tier operating policy (no true price required):

Tier	Action	Share	Median APE	Hit ≤20%
A — auto-serve	return value automatically	40%	9.55%	75.0%
B — serve with range	return value with P10–P90 band	40%	17.64%	54.3%
C — flag for review	route to human review	20%	28.85%	38.7%

Table 5: Sharp where confident, cautious where not. A complementary check: when a listing's asking price falls *inside* its interval (76.7% of listings) Median APE is 10.9%; when it falls *outside* it is 40.0%.

The dangerous failure mode of an AVM is not "sometimes wrong" — every AVM is — but "*confidently* wrong without warning." This system avoids that: its error is concentrated in a population it can identify up front.

5 Diving Into Each Region

Every province with ≥ 100 test listings has its own CSV file named after it (e.g. `regions/ho chi minh.csv`) holding all its listings with actual price, prediction, interval, percentage error, and a `case_type` label, sorted worst-error first. The 48 reported provinces cover ∼ 99% of listings.

All metrics are reported for every region, split across two tables for readability: Table 6 covers **accuracy** (sample size, MAE, MAPE, median APE, and the hit rates within 10% and 20%), and Table 7 covers **calibration and confidence** (raw / median / trimmed bias, mean interval width, in-band coverage, and the high-confidence median APE). Both are sorted by sample size.

Table 6: Accuracy metrics per region (production shrinkage-blend model).

Region	n	MAE (B)	MAPE%	Med.APE%	Hit10%	Hit20%
Da Nang	26,787	2.40	21.0	11.3	45.9	69.5
Ho Chi Minh	14,904	5.12	48.0	23.6	24.4	43.7
Hai Phong	13,952	1.57	23.8	12.8	42.3	65.1
Binh Duong	13,663	1.19	30.5	12.8	42.8	64.0
Dong Nai	12,978	1.70	32.0	14.6	39.5	59.2
Khanh Hoa	8,034	2.66	24.8	11.2	47.1	66.5
Ba Ria Vung Tau	6,873	2.79	27.9	11.9	45.5	64.6
Long An	6,496	1.07	33.5	17.7	33.4	53.7
Can Tho	3,514	2.40	30.0	13.3	42.3	62.1
Ha Noi	3,358	6.52	152.5	20.4	28.6	49.3
Hue	2,566	1.91	25.6	11.9	44.9	66.0
Dak Lak	2,228	1.45	32.1	14.0	39.6	62.2
Lam Dong	2,175	3.98	36.0	18.2	33.2	52.5
Bac Ninh	2,061	2.87	33.4	14.5	39.4	60.1
Quang Nam	1,896	3.55	32.0	17.6	33.0	54.4
Quang Ninh	1,565	5.38	45.8	18.0	33.1	53.2
Binh Thuan	1,518	3.12	39.2	20.0	31.7	49.9
Tay Ninh	1,166	1.20	35.7	22.1	29.8	46.4
Thanh Hoa	1,150	2.46	42.5	20.6	32.4	49.0
other small (46)	936	3.25	110.6	53.1	10.6	21.6
Hoa Binh	912	2.04	44.2	23.0	27.2	43.9
Nghe An	875	2.43	40.8	15.1	37.7	59.7
Vinh Phuc	843	2.38	39.4	18.0	33.9	54.0
Binh Dinh	836	1.98	28.4	14.4	38.8	60.4
Nam Dinh	815	1.11	23.0	13.7	39.6	63.7
Thai Binh	813	1.27	28.1	15.6	35.8	58.3
Hai Duong	712	2.60	29.6	13.8	40.9	60.5
Binh Phuoc	663	1.24	44.7	27.1	23.1	39.7
Bac Giang	582	1.90	44.8	26.4	22.3	41.1
Tien Giang	560	1.51	38.5	21.6	30.7	48.0
Kien Giang	531	5.02	55.7	26.9	25.8	42.6
Hung Yen	503	3.59	33.5	17.8	32.2	53.1
Phu Yen	454	1.65	36.0	19.6	29.3	50.4
Ha Nam	448	2.17	38.1	17.7	33.0	52.7
Tra Vinh	331	0.87	34.9	22.5	22.7	45.6
Ninh Thuan	323	1.53	55.3	20.5	32.5	49.2
Ben Tre	318	1.94	42.3	22.7	29.6	45.6
Thai Nguyen	304	1.75	40.2	15.8	36.5	58.2
Quang Ngai	303	1.08	42.4	20.2	29.4	49.8
Phu Tho	284	1.83	49.6	24.0	25.7	44.4
Gia Lai	238	1.69	49.8	25.8	24.4	45.8
Hau Giang	211	0.99	39.3	20.6	25.6	48.8
Lao Cai	208	7.06	88.5	29.5	21.6	38.9
An Giang	202	2.26	55.5	31.8	17.8	36.1
Dong Thap	180	1.44	46.8	31.0	23.3	37.8
Quang Tri	179	1.25	46.1	26.4	21.2	42.5
Quang Binh	165	2.58	37.0	18.0	30.9	54.5
Vinh Long	159	2.56	56.9	33.8	13.2	31.4
Son La	102	3.67	74.1	42.9	15.7	24.5

Table 7: Calibration and confidence per region. Median/trimmed bias near 0 = the model is centered.

Region	Raw Bias%	Med.Bias%	Trim.Bias%	Int.Width%	In-Band%	HiConf MedAPE%
Da Nang	5.3	-0.1	0.4	69	77	8.8
Ho Chi Minh	22.6	-0.7	3.3	130	76	18.6

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Table 7 – continued

Region	Raw Bias%	Med.Bias%	Trim.Bias%	Int.Width%	In-Band%	HiConf MedAPE%
Hai Phong	5.8	-0.7	-0.2	96	78	9.4
Binh Duong	13.5	-0.2	1.3	92	78	8.6
Dong Nai	12.5	-0.2	1.6	110	77	10.3
Khanh Hoa	7.7	-0.1	0.1	95	76	7.9
Ba Ria Vung Tau	9.5	-0.5	0.0	101	77	7.7
Long An	11.9	-0.0	2.6	144	78	12.7
Can Tho	11.2	-0.5	0.3	108	77	9.2
Ha Noi	127.7	-2.0	-0.6	107	76	18.1
Hue	8.9	-0.0	0.8	114	78	9.1
Dak Lak	13.8	-0.3	0.8	118	75	10.8
Lam Dong	14.2	-1.0	2.2	149	76	14.1
Bac Ninh	11.8	-1.1	-1.1	116	78	11.6
Quang Nam	9.0	-0.4	1.1	162	77	15.1
Quang Ninh	21.5	-0.6	0.1	139	74	14.7
Binh Thuan	12.9	-0.8	1.3	196	77	16.9
Tay Ninh	11.1	1.4	4.6	258	75	18.1
Thanh Hoa	19.3	1.5	5.0	199	75	12.4
other small (46)	78.4	15.6	40.0	218	70	39.6
Hoa Binh	18.5	1.3	5.9	265	75	20.0
Nghe An	19.9	-1.5	-0.5	118	79	12.5
Vinh Phuc	16.0	-1.6	-0.2	134	79	12.7
Binh Dinh	8.3	-1.5	-0.6	115	81	11.4
Nam Dinh	5.0	0.2	1.2	122	80	10.9
Thai Binh	6.7	-1.8	0.5	122	76	13.3
Hai Duong	5.4	-2.5	-2.5	118	77	9.1
Binh Phuoc	20.8	5.5	12.3	403	72	25.3
Bac Giang	14.2	0.3	1.9	222	73	21.5
Tien Giang	10.2	-0.9	0.5	226	79	16.1
Kien Giang	28.3	1.2	7.3	232	71	21.8
Hung Yen	9.1	-1.8	-0.6	138	78	11.9
Phu Yen	12.5	-0.8	1.1	180	76	19.3
Ha Nam	14.0	1.0	1.0	137	72	12.9
Tra Vinh	10.8	2.2	3.5	254	84	24.0
Ninh Thuan	35.1	2.4	10.1	221	76	17.8
Ben Tre	19.5	6.7	10.8	318	76	19.0
Thai Nguyen	16.8	-0.9	-0.1	168	77	11.8
Quang Ngai	23.8	5.5	8.1	207	77	16.2
Phu Tho	23.2	-0.9	4.6	201	70	19.5
Gia Lai	27.0	6.4	11.2	374	71	17.7
Hau Giang	18.9	10.9	11.2	524	65	28.4
Lao Cai	56.0	-3.8	9.0	163	69	22.9
An Giang	29.5	8.1	13.5	259	75	21.2
Dong Thap	21.9	4.8	11.8	266	71	21.6
Quang Tri	21.9	3.8	9.1	296	75	18.2
Quang Binh	9.2	-4.8	-2.7	153	70	11.6
Vinh Long	27.6	11.6	13.4	289	71	25.0
Son La	27.5	-9.7	2.7	212	72	33.2

How to read these tables (the three key insights).

1. **Accuracy is not uniform — and the report says so.** Well-sampled markets are sharp (Da Nang 11.3% median APE, Khanh Hoa 11.2%, Hai Phong / Binh Duong $\sim$ 13%), while **Ho Chi Minh is the clear weak spot** (23.7%) — a high-variance market the blended national average was hiding.

2. **Every region is unbiased.** In Table 7 the median and trimmed bias sit near 0 for essentially every province: no market is systematically over- or under-valued. The large raw MAPE values are the label-noise tail (Section 6), not a regional tilt.

3. **Confidence tracks difficulty.** Regions with wider mean intervals are exactly the higher-error regions, and the *high-confidence* median APE column is markedly sharper than the overall one in every region — the model's uncertainty signal works locally, not just in aggregate.

6 Error Cases per Region, and a Focus on the Worst

For every province, the worst error cases sit at the top of its CSV file. Ranking provinces by the share of large-error listings (absolute percentage error $> 50\%$) and comparing the price/m^2 of their error listings against their normal listings reveals a single, consistent cause.

Region	n	Large-error rate	Err. median ppm2	Normal median ppm2
Ho Chi Minh	14,904	21.2%	39.4M	77.6M
Long An	6,496	17.8%	7.5M	18.8M
Thanh Hoa	1,150	22.1%	15.0M	41.0M
Binh Thuan	1,518	20.8%	12.3M	27.8M

Table 8: In every region, error listings have a far *lower* price/m^2 than normal listings — the signature of underpriced / mislabeled listings, not model failure.

The error is in the label, not the model — shown for every region. Table 9 lists, for *each* region, its single worst-error listing next to a representative normal listing, together with the features that explain the gap (area, floors, price/m^2), the listed price, and the model's prediction. The table is meant to be read row-pair by row-pair, and every pair tells the same story:

- **The discriminating feature is price/m^2.** The worst-error listing always has an *implausibly low* price/m^2 (often 1–4M, versus tens of millions for the normal listing in the same region). Area and floors are ordinary — it is the price feature that is broken.
- **The prediction comparison proves the point.** For the worst case the *model's* value is far above the tiny listed price (e.g. tens or hundreds of billions vs. a fraction of a billion), because the model reconstructs a sensible value from the normal features. For the normal case the model's value sits within a few percent of the listed price. The huge "error%" is therefore the distance between a good prediction and a bad label — not model failure.

Table 9: Worst-error vs. a normal listing for every region, with the features that drive the gap. Error listings consistently show an implausibly low price/m^2; the model predicts a sensible value, so the large "error" reflects a bad listed price, not a bad prediction.

Region	Case	Subtype	Area	Flr	ppm2(M)	Listed(B)	Model(B)	Err%
Da Nang	worst	normal	82	4	1.2	0.10	9.81	9,706
Da Nang	normal	alley frontage	186	3	55.9	10.39	11.10	7
Ho Chi Minh	worst	frontage	160	10	1.5	0.24	127.56	53,050
Ho Chi Minh	normal	normal	56	–	89.3	5.00	5.44	9
Hai Phong	worst	frontage	68	2	1.5	0.10	8.74	8,642
Hai Phong	normal	alley frontage	100	5	108.0	10.80	11.56	7
Binh Duong	worst	normal	220	5	1.0	0.23	13.25	5,659
Binh Duong	normal	normal	64	3	52.3	3.35	3.57	7
Dong Nai	worst	frontage	99	–	1.0	0.10	7.22	7,120
Dong Nai	normal	normal	80	2	57.5	4.60	4.90	6
Khanh Hoa	worst	alley frontage	70	–	1.6	0.11	6.26	5,587
Khanh Hoa	normal	frontage	520	–	180.8	94.00	99.30	6
Ba Ria Vung Tau	worst	frontage	117	–	1.4	0.16	10.19	6,269
Ba Ria Vung Tau	normal	alley	54	2	72.2	3.90	4.13	6
Long An	worst	townhouse project	430	–	1.8	0.77	12.61	1,539
Long An	normal	townhouse project	52	–	28.8	1.50	1.39	7
Can Tho	worst	normal	80	3	4.0	0.32	5.57	1,640

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Table 9 – continued

Region	Case	Subtype	Area	Flr	ppm2(M)	Listed(B)	Model(B)	Err%
Can Tho	normal	villa	400	–	25.0	10.00	10.65	6
Ha Noi	worst	townhouse project	290	8	1.0	0.30	126.41	42,035
Ha Noi	normal	alley frontage	39	4	111.5	4.35	4.71	8
Hue	worst	frontage	84	3	2.6	0.22	8.48	3,754
Hue	normal	frontage	110	3	54.5	6.00	6.37	6
Dak Lak	worst	frontage	122	2	1.0	0.12	10.48	8,490
Dak Lak	normal	normal	88	1	33.5	2.95	2.73	7
Lam Dong	worst	villa	500	–	1.3	0.65	17.88	2,655
Lam Dong	normal	frontage	93	5	263.4	24.50	22.65	8
Bac Ninh	worst	shophouse	100	–	1.1	0.11	9.66	8,683
Bac Ninh	normal	villa	180	–	66.7	12.00	11.22	7
Quang Nam	worst	normal	127	–	2.8	0.35	3.19	810
Quang Nam	normal	townhouse project	160	–	15.6	2.50	2.31	8
Quang Ninh	worst	villa	296	–	1.0	0.30	35.23	11,645
Quang Ninh	normal	normal	2100	27	314.3	660.00	706.55	7
Binh Thuan	worst	townhouse project	75	–	2.7	0.20	3.37	1,587
Binh Thuan	normal	frontage	100	–	24.0	2.40	2.57	7
Tay Ninh	worst	frontage	174	–	1.3	0.22	1.44	554
Tay Ninh	normal	normal	232	–	8.6	2.00	1.85	7
Thanh Hoa	worst	frontage	35	3	3.7	0.13	2.67	1,953
Thanh Hoa	normal	frontage	637	4	45.5	29.00	27.09	7
Hoa Binh	worst	frontage	436	2	1.3	0.55	4.49	716
Hoa Binh	normal	villa	180	–	9.1	1.63	1.51	8
Nghe An	worst	alley frontage	79	4	2.2	0.18	13.12	7,354
Nghe An	normal	townhouse project	128	–	30.5	3.90	4.18	7
Vinh Phuc	worst	frontage	78	4	2.3	0.18	4.91	2,625
Vinh Phuc	normal	frontage	95	3	121.1	11.50	10.66	7
Binh Dinh	worst	normal	350	–	1.4	0.50	9.24	1,748
Binh Dinh	normal	frontage	42	2	57.1	2.40	2.58	7
Nam Dinh	worst	alley	469	–	1.5	0.70	6.84	876
Nam Dinh	normal	alley frontage	40	3	39.5	1.58	1.46	8
Thai Binh	worst	villa	240	–	2.6	0.62	4.17	568
Thai Binh	normal	normal	40	3	38.8	1.55	1.66	7
Hai Duong	worst	townhouse project	200	–	2.9	0.58	5.81	901
Hai Duong	normal	alley frontage	41	3	59.8	2.45	2.62	7
Binh Phuoc	worst	frontage	160	–	2.4	0.39	2.52	547
Binh Phuoc	normal	alley frontage	300	–	1.2	0.36	0.39	8
Bac Giang	worst	townhouse project	270	–	3.7	1.00	7.05	605
Bac Giang	normal	townhouse project	150	5	100.0	15.00	13.65	9
Tien Giang	worst	alley frontage	160	–	2.1	0.33	1.91	480
Tien Giang	normal	frontage	151	1	27.8	4.20	3.89	7
Kien Giang	worst	villa	13	–	8.0	0.10	1.20	1,085
Kien Giang	normal	normal	114	–	17.5	2.00	2.17	8
Hung Yen	worst	normal	62	–	2.8	0.17	2.17	1,149
Hung Yen	normal	normal	63	–	55.6	3.50	3.77	8
Phu Yen	worst	townhouse project	1169	–	2.0	2.35	14.80	530
Phu Yen	normal	frontage	100	–	41.0	4.10	3.76	8
Ha Nam	worst	townhouse project	165	–	1.7	0.28	10.87	3,844
Ha Nam	normal	townhouse project	52	5	101.0	5.25	4.88	7
Tra Vinh	worst	alley	485	–	1.0	0.48	3.84	691
Tra Vinh	normal	frontage	76	1	59.2	4.50	4.05	10
Ninh Thuan	worst	shophouse	210	4	2.3	0.48	5.66	1,086
Ninh Thuan	normal	shophouse	220	4	13.6	3.00	3.21	7
Ben Tre	worst	alley	80	–	3.1	0.25	1.33	430
Ben Tre	normal	normal	120	–	4.6	0.55	0.60	8
Thai Nguyen	worst	frontage	131	1	1.3	0.17	3.37	1,931
Thai Nguyen	normal	normal	40	2	50.0	2.00	1.86	7
Quang Ngai	worst	alley frontage	133	–	1.2	0.16	1.69	959
Quang Ngai	normal	frontage	105	3	31.0	3.25	3.51	8
Phu Tho	worst	villa	197	–	2.5	0.49	4.07	721
Phu Tho	normal	frontage	700	–	9.1	6.40	6.97	9
Gia Lai	worst	townhouse project	300	–	1.6	0.47	3.20	584
Gia Lai	normal	frontage	150	2	19.0	2.85	2.59	9
Hau Giang	worst	frontage	683	–	1.2	0.80	4.09	412
Hau Giang	normal	villa	138	–	12.0	1.65	1.80	9
Lao Cai	worst	townhouse project	120	4	1.0	0.12	4.16	3,368

continued on next page

Table 9 – continued

Region	Case	Subtype	Area	Flr	ppm2(M)	Listed(B)	Model(B)	Err%
Lao Cai	normal	frontage	145	–	51.0	7.40	6.75	9
An Giang	worst	townhouse project	200	–	3.0	0.59	4.08	591
An Giang	normal	frontage	180	–	34.4	6.20	5.58	10
Dong Thap	worst	normal	150	–	2.7	0.41	1.93	371
Dong Thap	normal	frontage	184	–	6.0	1.10	1.02	8
Quang Tri	worst	townhouse project	325	–	1.3	0.44	3.24	641
Quang Tri	normal	normal	142	–	5.3	0.76	0.68	10
Quang Binh	worst	townhouse project	200	–	5.0	1.00	4.97	397
Quang Binh	normal	townhouse project	50	–	42.0	2.10	2.28	9
Vinh Long	worst	frontage	361	–	1.0	0.37	4.11	1,010
Vinh Long	normal	shophouse	44	–	15.2	0.66	0.74	12
Son La	worst	villa	190	–	1.9	0.37	3.00	710
Son La	normal	townhouse project	85	4	51.8	4.40	4.76	8

Focus: Ho Chi Minh City — the region with the most errors

Ho Chi Minh is both the largest major market (14,904 test listings) and the one producing the most error cases (21.2% large-error rate, $\approx$ 3,160 listings). Its defensible interpretation:

- **The worst case is a data error, and the model caught it.** A 160 m^2 frontage house listed at 1.5M/m^2 (0.24B total) is impossible in Ho Chi Minh, where normal frontage houses trade near 80–100M/m^2. The model valued it at $\sim$ 128B — almost certainly the true order of magnitude — producing a 53,000% "error" that is entirely the fault of a mistyped price. This listing is exactly what an AVM should *flag*, not what it should be judged for missing.
- **Error listings are systematically cheap.** Their median price/m^2 is 39.4M versus 77.6M for normal listings — error cases are dominated by implausibly under-priced posts, and 19.9% sit in the extreme-price tail versus 8.8% of normal listings.
- **Genuine market variance, honestly reported.** Beyond bad labels, Ho Chi Minh is a high-dispersion metro: prices for superficially similar listings vary widely by exact street and alley. This is why its intervals are wider and more of its listings route to Tier C. The right response is district-level modeling, and the segmentation makes that need visible instead of averaging it away.
- **The product remains safe here.** On the higher-confidence half of Ho Chi Minh listings the typical error is $\sim$ 18%, and the anomaly checks (interval width, asking price outside band) route the problematic listings to review. Normal Ho Chi Minh listings — such as the 56 m^2 home valued to 8.8% above — are served sharply.

7 Limitations and Next Steps

- **Thin regions stay uncertain.** Small provinces have wider intervals and larger errors; the policy correctly routes more of them to Tier C. More regional data is the only real fix.
- **The error tail is contained, not cleaned.** Flagging limits the damage; cleaning the underlying suspicious-price listings is the upstream fix and the single biggest lever on raw MAPE.
- **Random split.** All numbers use a random test split; a time-based split (hold out the most recent months) is the next step to prove the model holds on *future* listings.
- **Ho Chi Minh** is the clearest modeling follow-up — a high-variance metropolitan market that likely needs district-level segmentation.

All figures are computed from the held-out test predictions of the production (shrinkage-blend) model. Per-region data lives in `Rule-based Scene/Evaluation/regional_report/regions/`; the full machine-readable report is regenerated with `python3 src/build_regional_report.py`.